



LNG Cargo Diversion Decision Engine

Structured decision framework for Europe vs Asia LNG diversion economics.

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Executive Summary

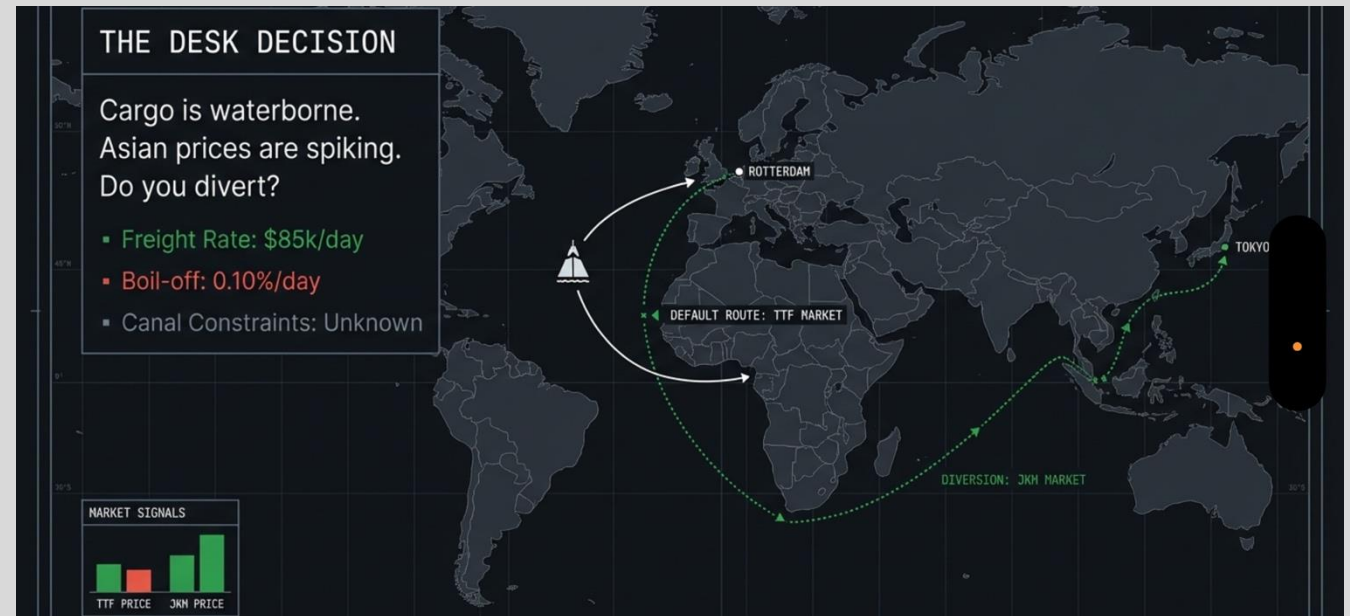
Problem

Given an LNG cargo loading in the US Gulf

Should the vessel discharge in Europe (TTF) or divert to Asia (JKM)?

Decision requires integrating:

- Price spreads
- Freight economics
- Carbon costs
- Operational risk buffers
- Hedge execution considerations



Approach

A rule-based decision engine that:

- Calculates netback economics
- Applies conservative risk adjustments
- Generates executable hedge instructions
- Validates behavior via historical rule testing

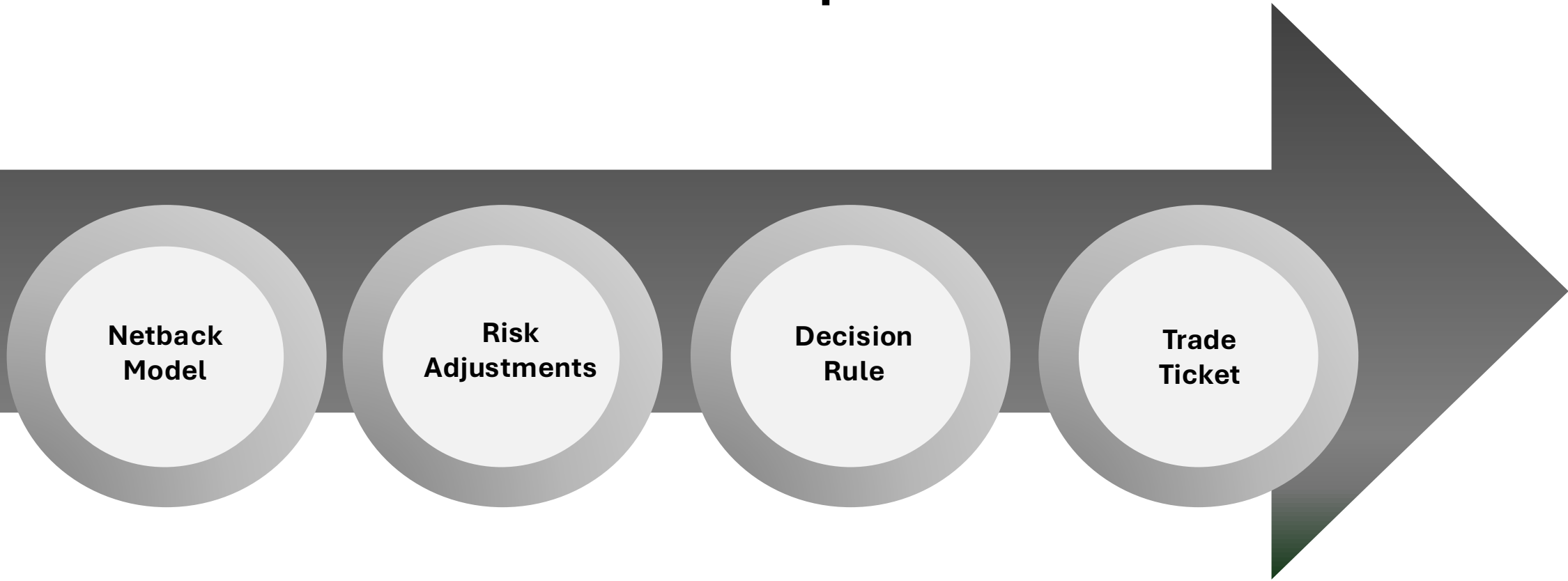
Example Output

DIVERT TRIGGERED	
Europe Netback	\$126.26 M
Asia Netback	\$133.96 M
Adjusted Uplift	\$7.07 M
Hedge	BUY JKM 322 lots / SELL TTF 322 lots

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Decision Framework

Decision Framework - Market Inputs

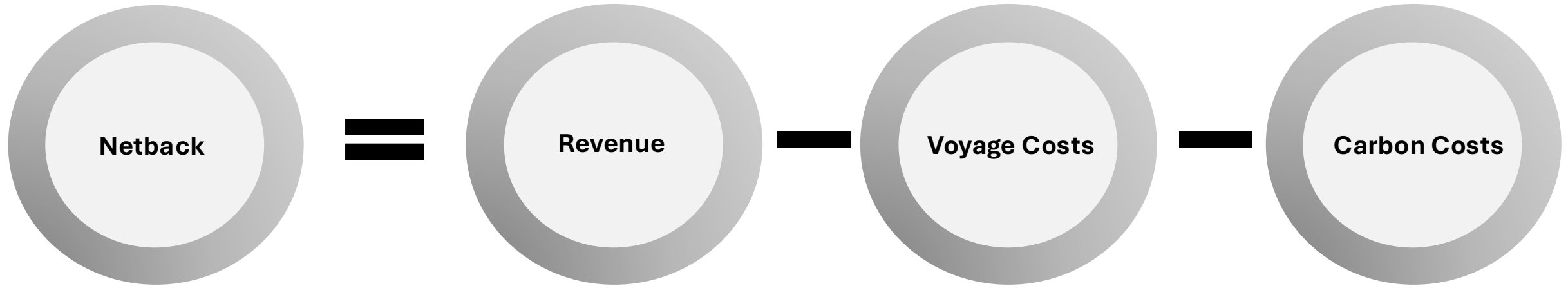


- Raw uplift = Asia netback – Europe netback
- Basis haircut applied
- Operational buffer deducted
- Decision triggered only above threshold.

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Netback Methodology

Netback Calculation Logic



- Distance ➡ Voyage time
- Revenue = Destination Price x Delivered LNG Energy
- Fuel consumption
- Boil-off adjustment
- EUA exposure
- Delivered energy

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Example Diversion Decision

Example Trade Output

1. Clear decision
2. Hedge sizing
3. Adjusted uplift logic

LNG DIVERSION TRADE NOTE

Route: US_Gulf → Rotterdam (TTF) vs Tokyo (JKM)
Vessel: TFDE | Cargo: 174,000 m³
As of: 2026-02-23 00:31 UTC

TTF: \$32.03/MMBtu (real)
JKM: \$34.78/MMBtu (proxy)
Freight: \$85,000/day (proxy)
Fuel: \$747/t (proxy)
EUA: \$69.21/tCO₂ (real)

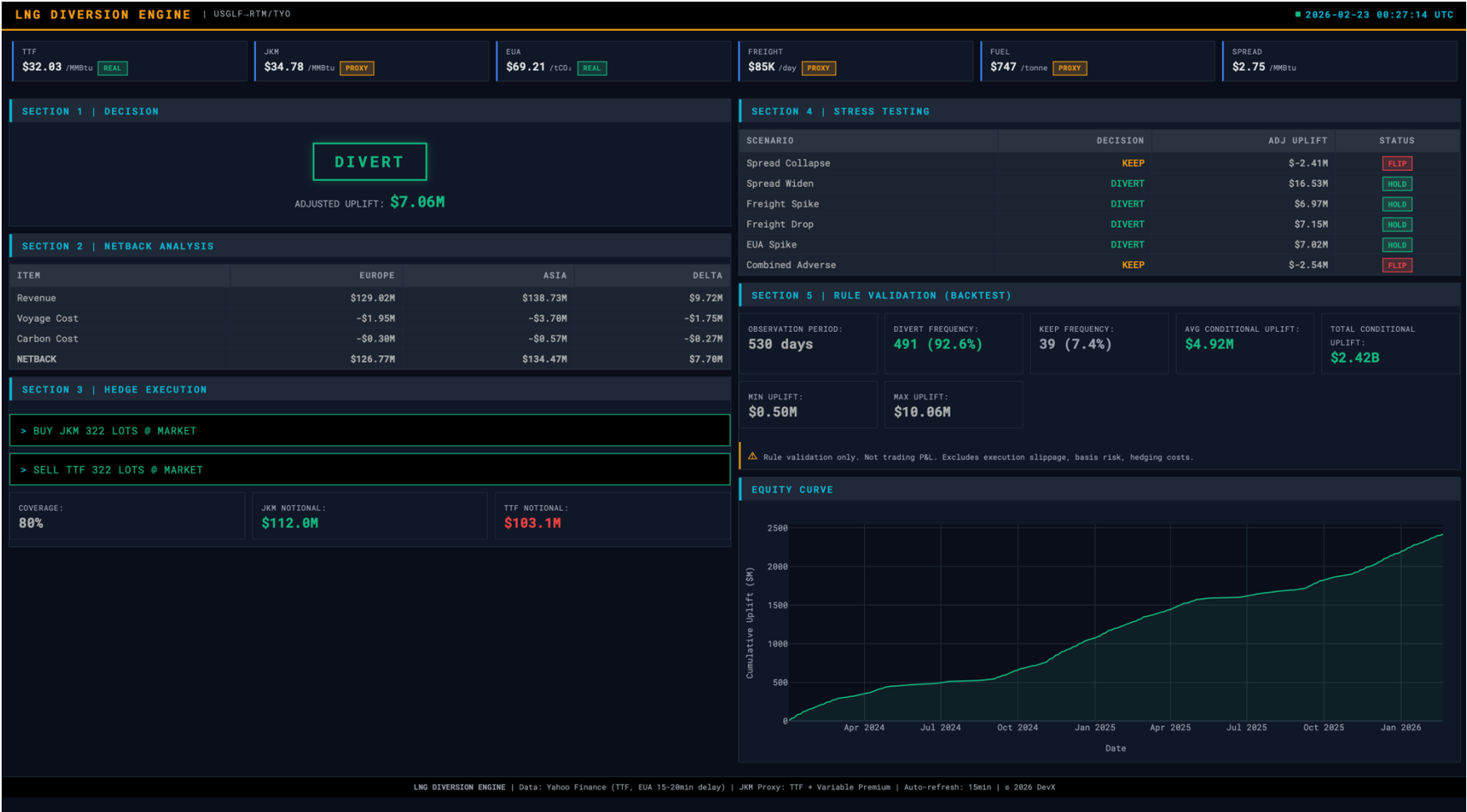
Europe netback: \$126,771,038
Asia netback: \$134,466,238
Raw uplift: \$7,695,200
Adjusted uplift: \$7,060,440
(basis=5.0%, ops=\$250,000, threshold=\$500,000)

✓ Decision: DIVERT

Hedge: BUY JKM 322 lots | SELL TTF 322 lots

Terminal Dashboard - LNG Diversion Engine

- 1. Route: US Gulf to Rotterdam vs Tokyo
- 2. Adjusted uplift: \$7.06M
- 3. Decision: DIVERT
- 4. Hedge: BUY JKM / SELL TTF



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Risk & Stress Testing

Decision Robustness

SCENARIO	DECISION	ADJ UPLIFT	STATUS
Spread Collapse	KEEP	\$-2.41M	FLIP
Spread Widen	DIVERT	\$16.53M	HOLD
Freight Spike	DIVERT	\$6.97M	HOLD
Freight Drop	DIVERT	\$7.15M	HOLD
EUA Spike	DIVERT	\$7.02M	HOLD
Combined Adverse	KEEP	\$-2.54M	FLIP

- Decision flips under spread collapse
- Demonstrates sensitivity to freight/carbon shocks

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Rule Behaviour (Historical Validation)

Rule Behaviour Over Time

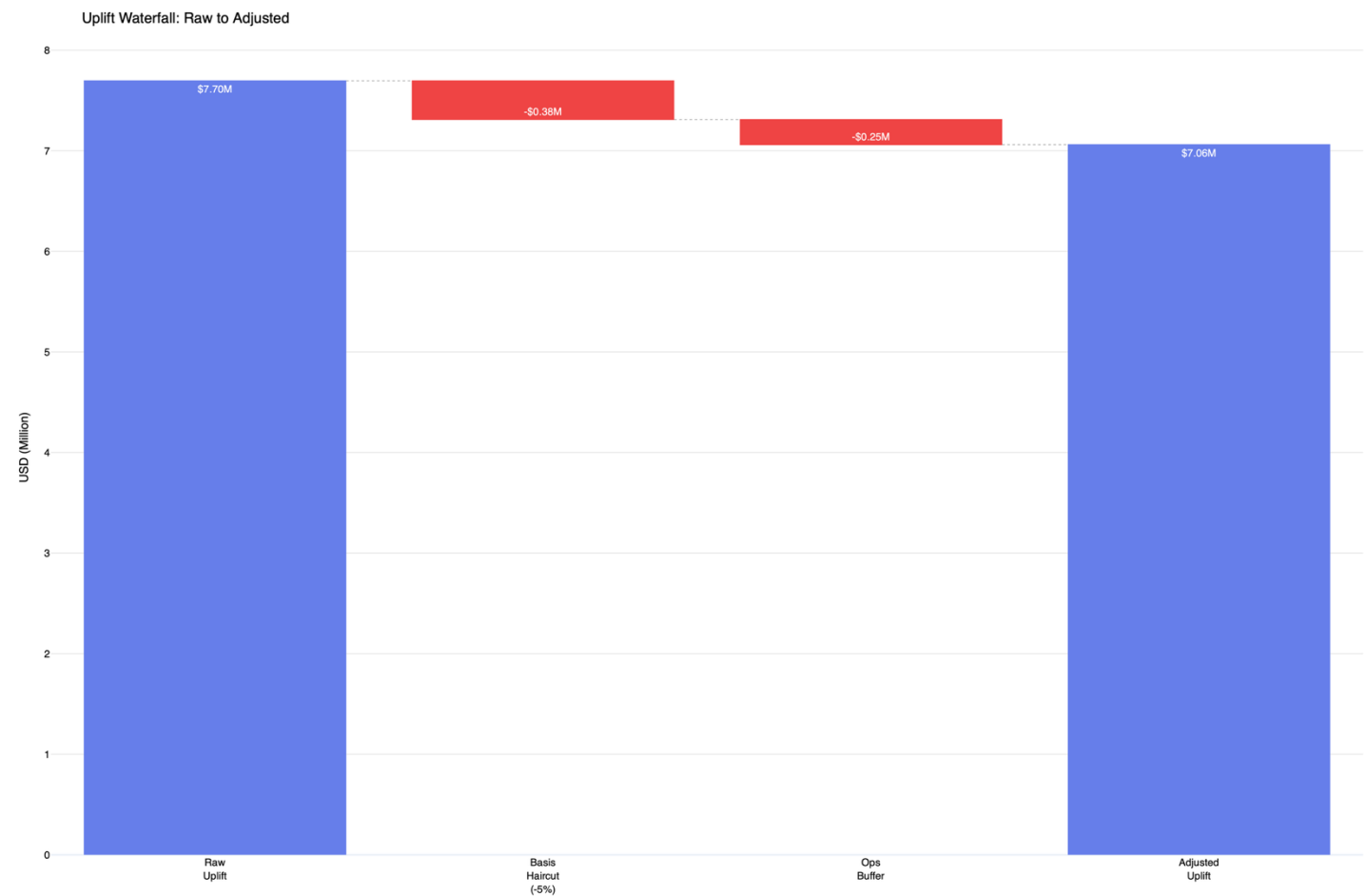


Disclaimer: Historical simulation indicates diversion conditions frequently satisfied under proxy spread dynamics

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Decision Waterfall

From Raw Spread to Executable Diversion Decision



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Assumptions & Next Steps

Model Limitations and Future Enhancements

Limitations	Next Evolution
JKM Proxy	Licensed JKM feed
Freight Proxy	Freight curve integration
Simplified Routing	Stochastic ops delays
No demurrage modelling	Optionality modelling
Contract structure (FOB vs DES) not modelled	Contract Aware diversion logic